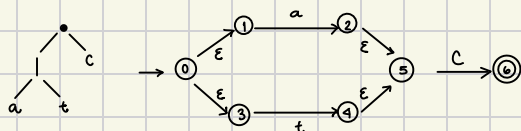
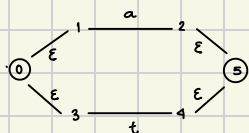


→ Laboratorio #2 ←

a) $(a|t)c$



$$\Rightarrow \Sigma = \{a, t, c\}$$

$$s_0 = 0$$

$$F = \{6\}$$

Tabla de transición del AFN

	a	t	c	ϵ^*
0	-	-	-	$\{0, 1, 3\}$
1	2	-	-	$\{1\}$
2	-	-	-	$\{2, 5\}$
3	-	4	-	$\{3\}$
4	-	-	-	$\{4, 5\}$
5	-	-	6	$\{5\}$
6	-	-	-	$\{6\}$

Estados del AFN

$$A = \{0, 1, 3\}$$

$$B = \{2, 5\}$$

$$C = \{4, 5\}$$

$$D = \{6\}$$

$$Q = \{A, B, C, D\}$$

$a\epsilon^*$

$$\{2, 5\} = B$$

$$-$$

$$-$$

$$-$$

$$Z = \{a, t, c\}$$

$t\epsilon^*$

$$\{4, 5\} = C$$

$$-$$

$$-$$

$$-$$

$$q_0 = A$$

$c\epsilon^*$

$$\{6\} = D$$

$$-$$

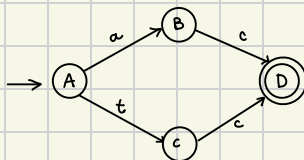
$$\{6\} = D$$

$$-$$

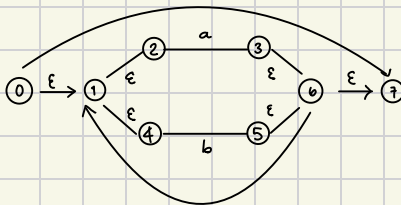
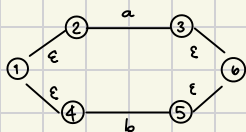
$$F = \{D\}$$

Diagrama del AFD

Estado AFN	a	t	c
$\xrightarrow{\text{inicio}}$ A	B	C	-
B	-	-	D
C	-	-	D
D	-	-	-



Ejercicio b) $(a|b)^*$



final.

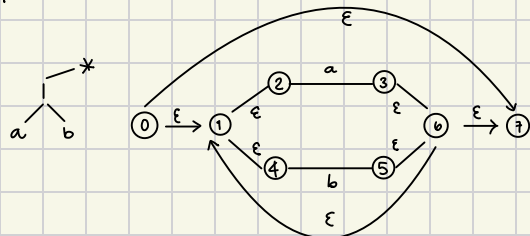


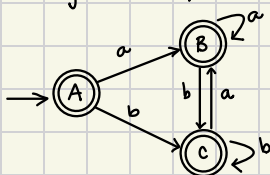
Tabla de transición AFN

	a	b	ϵ^*
Inicio $\rightarrow$ 0	/	/	$\{0, 1, 2, 4, 7\}$
1	/	/	$\{1, 2, 4\}$
2	3	/	$\{2\}$
3	/	/	$\{3, 6, 1, 2, 4, 7\}$
4	/	5	$\{4\}$
5	/	/	$\{5, 6, 7, 1, 2, 4\}$
6	/	/	$\{1, 2, 4, 5, 6, 7\}$
7	/	/	$\{6, 7, 1, 2, 4\}$
			$\{1, 2, 4, 6, 7\}$
			$\{7\}$

Estados del AFN	$a\epsilon^*$	$b\epsilon^*$
$\{0, 1, 2, 4, 7\} = A$	$\{1, 2, 3, 4, 6, 7\} = B$	$\{1, 2, 4, 5, 6, 7\} = C$
$\{1, 2, 3, 4, 6, 7\} = B$	$\{1, 2, 3, 4, 6, 7\} = B$	$\{1, 2, 4, 5, 6, 7\} = C$
$\{1, 2, 4, 5, 6, 7\} = C$	$\{1, 2, 3, 4, 6, 7\} = B$	$\{1, 2, 4, 5, 6, 7\} = C$
A, B y C: aceptación		

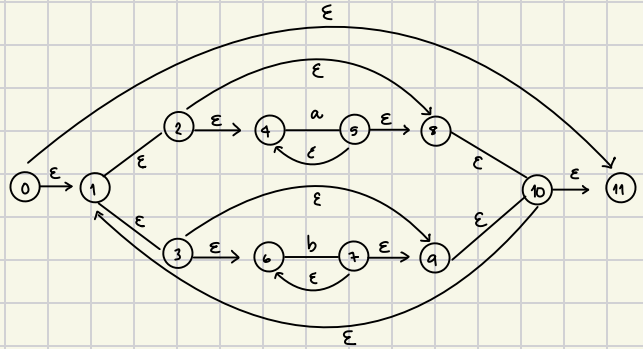
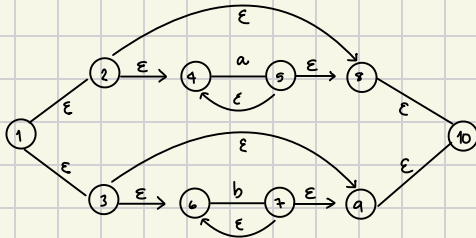
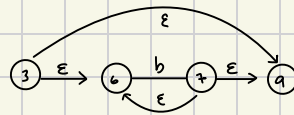
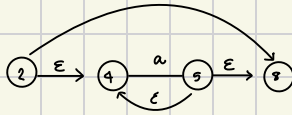
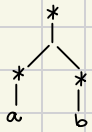
Estado AFN	a	b
Inicio $\rightarrow$ A	B	C
B	B	C
C	B	C

Diagrama del AFD

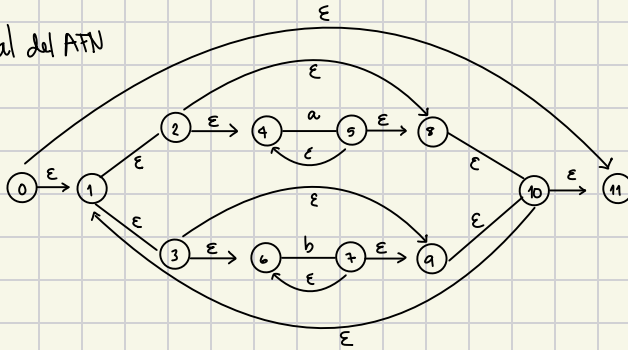
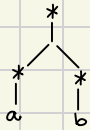


$$Q = \{A, B, C\} \quad \Sigma = \{a, b\} \quad q_0 = A \quad F = \{A, B, C\}$$

Ejercicio c) $(a^* | b^*)^*$



Respuesta final del AFN



Transición del AFN

	a	b	ϵ^*
0	/	/	$\{0, 1, 2, 3, 4, 6, 8, 9, 10, 11\}$
1	/	/	$\{1, 2, 3, 4, 6, 8, 9, 10, 11\}$
2	/	/	$\{1, 2, 3, 4, 6, 8, 9, 10, 11\}$
3	/	/	$\{1, 2, 3, 4, 6, 8, 9, 10, 11\}$
4	5	/	$\{4\}$
5	/	/	$\{1, 2, 3, 4, 5, 6, 8, 9, 10, 11\}$
6	/	7	$\{6\}$
7	/	/	$\{1, 2, 3, 4, 6, 7, 8, 9, 10, 11\}$
8	/	/	$\{1, 2, 3, 4, 6, 8, 9, 10, 11\}$
9	/	/	$\{1, 2, 3, 4, 6, 8, 9, 10, 11\}$
10	/	/	$\{1, 2, 3, 4, 6, 8, 9, 10, 11\}$
11	/	/	$\{11\}$

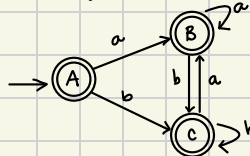
Estados del AFN

	$a\epsilon^*$	$b\epsilon^*$
$A = \{0, 1, 2, 3, 4, 6, 8, 9, 10, 11\}$	$B = \{1, 2, 3, 4, 5, 6, 8, 9, 10, 11\}$	$C = \{1, 2, 3, 4, 6, 7, 8, 9, 10, 11\}$
$B = \{1, 2, 3, 4, 5, 6, 8, 9, 10, 11\}$	$B = \{1, 2, 3, 4, 5, 6, 8, 9, 10, 11\}$	$C = \{1, 2, 3, 4, 6, 7, 8, 9, 10, 11\}$
$C = \{1, 2, 3, 4, 6, 7, 8, 9, 10, 11\}$	$B = \{1, 2, 3, 4, 5, 6, 8, 9, 10, 11\}$	$C = \{1, 2, 3, 4, 6, 7, 8, 9, 10, 11\}$

A, B y C: aceptación

Estado AFN	a	b
$\xrightarrow{\text{inicio}} A$	B	C
B	B	C
C	B	C

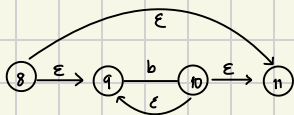
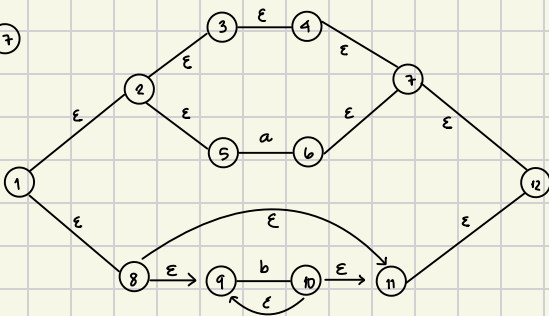
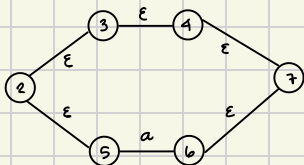
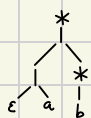
Diagrama del AFD



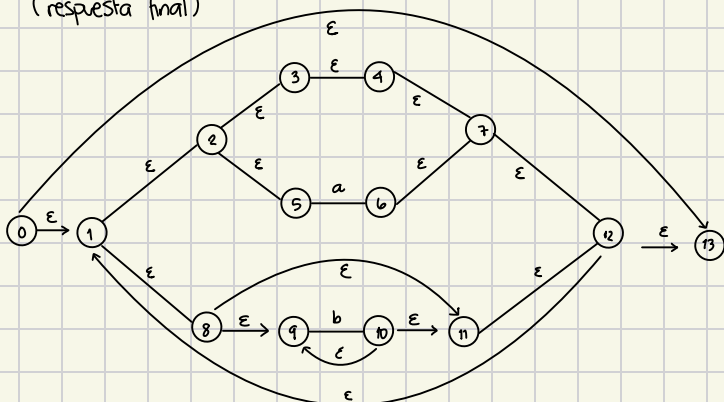
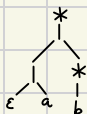
$$Q = \{A, B, C\} \quad \Sigma = \{a, b\} \quad q_0 = A \quad F = \{A, B, C\}$$

Problema d) $((\epsilon|a|b^*)^*)^*$

AFN:



7) nodo * a 6 (respuesta final)



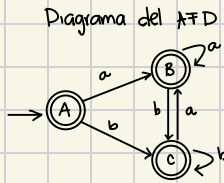
Transición del AFN

	a	b	ϵ^*
0	/	/	$\{0, 1, 2, 3, 4, 5, 7, 8, 9, 11, 12, 13\}$
1	/	/	$\{1, 2, 3, 4, 5, 7, 8, 9, 11, 12, 13\}$
2	/	/	$\{1, 2, 3, 4, 5, 7, 8, 9, 11, 12, 13\}$
3	/	/	$\{1, 2, 3, 4, 5, 7, 8, 9, 11, 12, 13\}$
4	/	/	$\{1, 2, 3, 4, 5, 7, 8, 9, 11, 12, 13\}$
5	6	/	$\{5\}$
6	/	/	$\{1, 2, 3, 4, 5, 6, 7, 8, 9, 11, 12, 13\}$
7	/	/	$\{1, 2, 3, 4, 5, 7, 8, 9, 11, 12, 13\}$
8	/	/	$\{1, 2, 3, 4, 5, 7, 8, 9, 11, 12, 13\}$
9	/	10	$\{9\}$
10	/	/	$\{1, 2, 3, 4, 5, 7, 8, 9, 10, 11, 12, 13\}$
11	/	/	$\{1, 2, 3, 4, 5, 7, 8, 9, 11, 12, 13\}$
12	/	/	$\{1, 2, 3, 4, 5, 7, 8, 9, 11, 12, 13\}$
13	/	/	$\{13\}$

Estados AFD

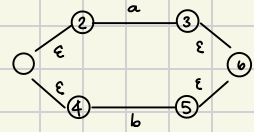
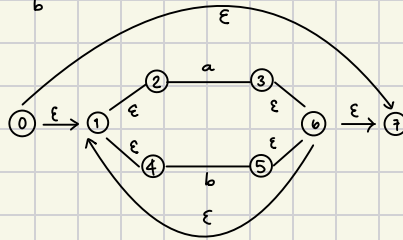
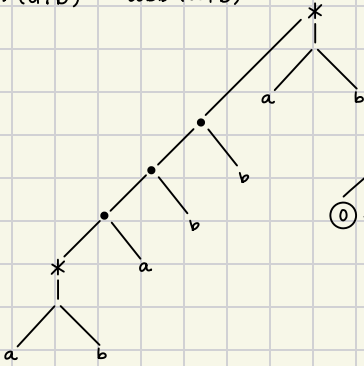
	a	b
$\{0, 1, 2, 3, 4, 5, 7, 8, 9, 11, 12, 13\} = A$	$\{1, 2, 3, 4, 5, 6, 7, 8, 9, 11, 12, 13\} = B$	$\{1, 2, 3, 4, 5, 7, 8, 9, 10, 12, 13\} = C$
$\{1, 2, 3, 4, 5, 6, 7, 8, 9, 11, 12, 13\} = B$	$\{1, 2, 3, 4, 5, 6, 7, 8, 9, 11, 12, 13\} = B$	$\{1, 2, 3, 4, 5, 7, 8, 9, 10, 12, 13\} = C$
$\{1, 2, 3, 4, 5, 7, 8, 9, 10, 11, 12, 13\} = C$	$\{1, 2, 3, 4, 5, 6, 7, 8, 9, 11, 12, 13\} = B$	$\{1, 2, 3, 4, 5, 7, 8, 9, 10, 12, 13\} = C$

Estado AFN	a	b
$\text{inicio} \rightarrow A$	B	C
B	B	C
C	B	C

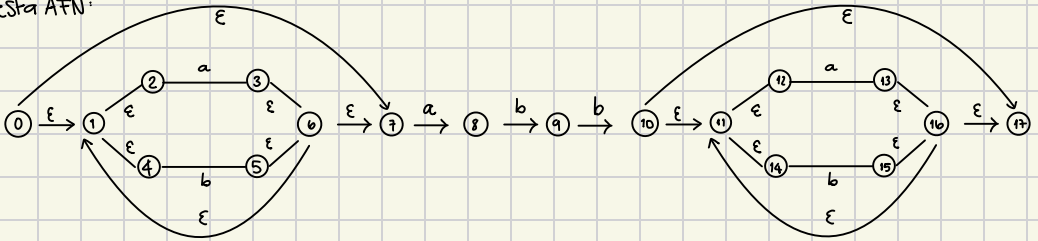


$Q = \{A, B, C\}$ $\Sigma = \{a, b\}$ $q_0 = A$ $F = \{A, B, C\}$

e) $(a|b)^* abb(a|b)^*$



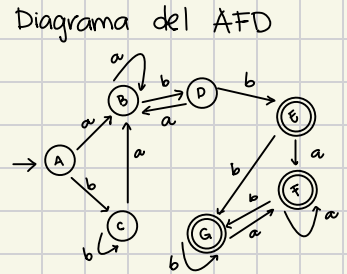
Respuesta AFN:



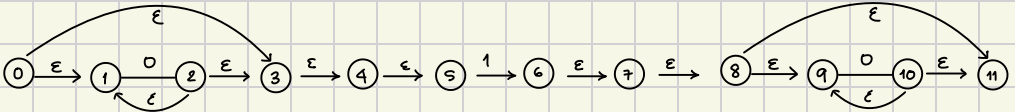
Estado	a	b	ϵ^*
0			$\{0, 1, 2, 4, 7\}$
1			$\{1, 2, 4\}$
2	3		$\{2\}$
3			$\{1, 2, 3, 4, 6, 7\}$
4		5	$\{4\}$
5			$\{1, 2, 4, 5, 6, 7\}$
6			$\{1, 2, 4, 6, 7\}$
7	8		$\{7\}$
8		9	$\{8\}$
9		10	$\{9\}$
10			$\{10, 11, 12, 14, 17\}$
11			$\{11, 12, 14\}$
12	13		$\{12\}$
13			$\{11, 12, 13, 14, 16, 17\}$
14		15	$\{14\}$
15			$\{11, 12, 14, 15, 16, 17\}$
16			$\{11, 12, 14, 16, 17\}$
17			$\{17\}$

Estado	a	b
$A = \{0, 1, 2, 4, 7\}$	$B = \{1, 2, 3, 4, 6, 7, 8\}$	$C = \{1, 2, 4, 5, 6, 7\}$
$B = \{1, 2, 3, 4, 6, 7, 8\}$	$B = \{1, 2, 3, 4, 6, 7, 8\}$	$D = \{1, 2, 4, 5, 6, 7, 9\}$
$C = \{1, 2, 4, 5, 6, 7\}$	$B = \{1, 2, 3, 4, 6, 7, 8\}$	$C = \{1, 2, 4, 5, 6, 7\}$
$D = \{1, 2, 4, 5, 6, 7, 9\}$	$B = \{1, 2, 3, 4, 6, 7, 8\}$	$E = \{1, 2, 4, 5, 6, 7, 10, 11, 12, 14, 17\}$
$(E) = \{1, 2, 4, 5, 6, 7, 10, 11, 12, 14, 17\}$	$F = \{1, 2, 3, 4, 6, 7, 8, 11, 12, 13, 14, 16, 17\}$	$G = \{1, 2, 4, 5, 6, 7, 11, 12, 14, 15, 16, 17\}$
$(F) = \{1, 2, 3, 4, 6, 7, 8, 11, 12, 13, 14, 16, 17\}$	$F = \{1, 2, 3, 4, 6, 7, 8, 11, 12, 13, 14, 16, 17\}$	$G = \{1, 2, 4, 5, 6, 7, 11, 12, 14, 15, 16, 17\}$
$(G) = \{1, 2, 4, 5, 6, 7, 11, 12, 14, 15, 16, 17\}$	$F = \{1, 2, 3, 4, 6, 7, 8, 11, 12, 13, 14, 16, 17\}$	$G = \{1, 2, 4, 5, 6, 7, 11, 12, 14, 15, 16, 17\}$

	a	b
inicio $\rightarrow$ A	B	C
B	B	D
C	B	C
D	B	E
E	F	G
F	F	G
G	F	G



Inicio f) 0? (1?) 0*

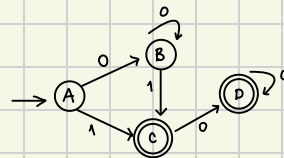


	0	1	ϵ
$\rightarrow 0$			$\{0, 1, 3, 4, 5\}$
1	2		$\{1\}$
2			$\{1, 2, 3, 4, 5\}$
3			$\{3, 4, 5\}$
4			$\{4, 5\}$
5	6		$\{5\}$
6			$\{6, 7, 8, 9, 11\}$
7			$\{7, 8, 9, 11\}$
8			$\{8, 9, 11\}$
9	10		$\{9\}$
10			$\{9, 10, 11\}$
11			$\{11\}$

transición AFD	0	1
$\rightarrow A = \{0, 1, 3, 4, 5\}$	$B = \{1, 2, 3, 4, 5\}$	$C = \{6, 7, 8, 9, 11\}$
$B = \{1, 2, 3, 4, 5\}$	$B = \{1, 2, 3, 4, 5\}$	$C = \{6, 7, 8, 9, 11\}$
$(C) = \{6, 7, 8, 9, 11\}$	$D = \{9, 10, 11\}$	—
$(D) = \{9, 10, 11\}$	$D = \{9, 10, 11\}$	—

transición AFD

	0	1
$\rightarrow A$	B	C
B	B	C
(C)	D	—
(D)	D	—



$Q = \{A, B, C, D\}$ $\Sigma = \{0, 1\}$ $q_0 = A$ $F = \{C, D\}$